An Introduction to Renormalization

From Divergent Integrals to Running Couplings

Particle Physics Reading Guide

Contents

1 The Problem: Infinities in Quantum Field Theory **2**
- 1.1 Where the infinities come from . . . 2
- 1.2 A concrete example: scalar field theory . . . 2
- 1.3 The one-loop correction to scattering . . . 3

2 The Strategy: What Renormalization Actually Does **3**
- 2.1 Parameters vs. observables . . . 3
- 2.2 The renormalization procedure, step by step . . . 4
- 2.3 An analogy: the broken altimeter . . . 4

3 Renormalization in ϕ^4 Theory: A Worked Example **4**
- 3.1 Regulating the bubble integral . . . 4
- 3.2 The total amplitude to one-loop order . . . 5
- 3.3 Defining the physical coupling . . . 5

4 Renormalization in QED: Vacuum Polarization **6**
- 4.1 The tree-level amplitude . . . 6
- 4.2 The vacuum polarization loop . . . 6
- 4.3 Tensor structure . . . 6
- 4.4 The modified propagator . . . 6
- 4.5 Evaluating $I(q^2)$: the divergence . . . 7
- 4.6 Defining the physical charge . . . 7

5 The Running Coupling Constant **7**
- 5.1 $\alpha(q^2)$ in QED . . . 7
- 5.2 How big is the effect? . . . 8
- 5.3 Contrast with QCD . . . 8

6 The Ward Identity: Why Charge Stays Universal **8**

7 What Makes a Theory Renormalizable? **9**
- 7.1 Power counting: a quick diagnostic . . . 9

8 The Renormalization Group: The Bigger Picture **10**
- 8.1 The β-function . . . 10
- 8.2 Fixed points . . . 10

9 Summary: The Logic of Renormalization **11**

1 The Problem: Infinities in Quantum Field Theory

Why This Matters

Quantum field theory (QFT) is spectacularly successful—QED predictions match experiment to 12 decimal places. But when you try to compute anything beyond the simplest ("tree-level") diagrams, you get integrals that diverge. Renormalization is the systematic procedure for extracting finite, physically meaningful predictions from these divergent expressions.

1.1 Where the infinities come from

Consider any QFT with interacting fields. At lowest order (tree level), everything is fine: you draw a Feynman diagram with no closed loops, write down the amplitude using the Feynman rules, and get a finite answer.

The trouble starts at *next* order. Whenever a Feynman diagram contains a closed loop, the internal momentum circulating around that loop is not fixed by external kinematics—it can take *any* value. You must integrate over all possible loop momenta:

$$\int \frac{d^4k}{(2\pi)^4} \text{ (stuff involving } k) \tag{1}$$

The integration runs from $k = 0$ all the way to $k \to \infty$. Depending on how the integrand behaves at large k (the "ultraviolet" region), this integral may diverge.

The Physical Source of Divergences

Physically, the divergence means: we're assuming the theory is valid at *arbitrarily short distances* (arbitrarily high momenta). That's almost certainly wrong—at some scale, new physics must enter (quantum gravity, string theory, something). The divergence is telling us we're asking the theory more than it can honestly answer about ultra-short-distance behavior.

1.2 A concrete example: scalar field theory

To build intuition without the complications of spinors and polarization vectors, consider the simplest interacting QFT: a real scalar field ϕ with a quartic self-interaction. The Lagrangian is:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \tag{2}$$

This is called ϕ^4 theory. It has one coupling constant λ and one mass m. The Feynman rules are simple:

- **Propagator** (internal scalar line carrying momentum k): $\dfrac{i}{k^2 - m^2}$
- **Vertex** (four lines meeting at a point): $-i\lambda$
- **Loop integral**: $\displaystyle\int \frac{d^4k}{(2\pi)^4}$ for each undetermined loop momentum

1.3 The one-loop correction to scattering

At tree level, two particles scatter via a single vertex: four external lines meet at one point, giving $\mathcal{M}_{\text{tree}} = -i\lambda$. Simple, finite, no problems.

At one-loop order, one of the corrections looks like this: the four-point vertex gets replaced by a "bubble"—two propagators forming a closed loop. The amplitude for this bubble diagram (in the s-channel) is:

$$\mathcal{M}_{\text{loop}} = \frac{(-i\lambda)^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2} \cdot \frac{i}{(k-p)^2 - m^2} \tag{3}$$

where p is the total external momentum flowing through the bubble, and the factor of $1/2$ is a symmetry factor.

Does This Integral Converge?

Count powers of k in the integrand. At large $|k|$:

Numerator: d^4k gives $k^3\,dk$ (in "spherical" coordinates in 4D momentum space).
Denominator: Two propagators $\sim 1/k^2 \times 1/k^2 = 1/k^4$.
Net behavior: $\displaystyle\int k^3\,dk \cdot \frac{1}{k^4} = \int \frac{dk}{k} = \ln k \Big|^{\infty} \to \infty$

It's logarithmically divergent.

This is the prototype of the problem. Every closed loop in every QFT potentially brings a divergent integral.

2 The Strategy: What Renormalization Actually Does

The Key Idea in One Sentence

We cannot compute absolute values of masses and couplings from first principles (they come out infinite), but we *can* compute how physical quantities *change* between different energy scales—and those changes are always finite.

2.1 Parameters vs. observables

A QFT Lagrangian contains **parameters**: masses m, coupling constants λ, g_e, etc. These are not the same as what you measure in the lab. What you measure are **observables**: scattering cross sections, decay rates, energy level shifts.

The connection between parameters and observables goes through the Feynman rules. At tree level, the connection is simple:

$$\lambda_{\text{measured}} \approx \lambda_{\text{Lagrangian}} \quad \text{(at lowest order)} \tag{4}$$

But at higher orders, loop corrections intervene. The "bare" parameters in the Lagrangian (m_0, λ_0) are *not* the physical quantities. They never were—we just didn't notice at tree level because the corrections hadn't shown up yet.

2.2 The renormalization procedure, step by step

The Procedure

1. **Regulate**: Introduce a temporary device (a "regulator") that makes all integrals finite. Common choices:

   - *Momentum cutoff* Λ: cap the loop integral at $|k| = \Lambda$.
   - *Dimensional regularization*: compute in $d = 4 - \epsilon$ dimensions, where integrals converge, then take $\epsilon \to 0$ at the end.

   The regulator is a crutch—the final physical answer must not depend on it.

2. **Identify the divergences**: With the regulator in place, compute the loop integrals. They now give finite but regulator-dependent answers (e.g., expressions containing $\ln \Lambda$ or $1/\epsilon$).

3. **Absorb into physical parameters**: Rewrite the Lagrangian parameters in terms of *physical* (renormalized) parameters:

$$\lambda_0 = \lambda_R + \delta\lambda, \qquad m_0^2 = m_R^2 + \delta m^2 \tag{5}$$

   where the "counterterms" $\delta\lambda$ and δm^2 contain exactly the right infinities to cancel the loop divergences.

4. **Express everything in terms of λ_R, m_R**: When you rewrite all amplitudes in terms of renormalized parameters, the regulator dependence cancels. You're left with finite, predictive expressions.

2.3 An analogy: the broken altimeter

Suppose your altimeter reads infinity for every mountain. Useless for absolute heights. But point it at two different mountains and take the *difference*—the broken offset cancels, and you get an accurate relative height.

Renormalization works the same way. The "absolute" values of m and λ come out infinite (broken altimeter). But *differences* between predictions at different energy scales are finite and physically meaningful.

3 Renormalization in ϕ^4 Theory: A Worked Example

3.1 Regulating the bubble integral

Return to the bubble integral of Eq. (3). Impose a hard momentum cutoff Λ:

$$\mathcal{M}_{\text{loop}}(s) = \frac{\lambda^2}{2} \int^{\Lambda} \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 - m^2)[(k-p)^2 - m^2]} \tag{6}$$

After doing the integral (using standard techniques like Feynman parameters), the result has the structure:

$$\mathcal{M}_{\text{loop}}(s) = i\lambda^2 \left[a \ln\left(\frac{\Lambda^2}{m^2}\right) + \text{finite function of } s \right] \tag{7}$$

where $s = p^2$ is the Mandelstam variable and a is a calculable numerical constant. The first term is the divergence (it blows up as $\Lambda \to \infty$); the second is perfectly finite.

Why the Divergence Doesn't Depend on s

The divergence comes from the region $|k| \gg |p|$, where the external momentum p is negligible compared to the loop momentum. In that regime, the integrand doesn't "know" what s is. So the infinite part is a pure constant—independent of external kinematics. This is the crucial property that makes renormalization possible.

3.2 The total amplitude to one-loop order

There are actually three bubble diagrams (one for each Mandelstam channel: s, t, u). Including all three plus the tree-level piece:

$$i\mathcal{M} = -i\lambda + i\lambda^2\Big[a\ln\Lambda^2 - b(s) - b(t) - b(u)\Big] \tag{8}$$

where $b(s)$, $b(t)$, $b(u)$ are finite functions of the kinematic variables, and I've absorbed various constants into a.

3.3 Defining the physical coupling

We *define* the physical coupling constant λ_R as the value of the scattering amplitude measured at some reference momentum configuration (s_0, t_0, u_0):

$$-i\lambda_R \equiv i\mathcal{M}\Big|_{s_0,t_0,u_0} = -i\lambda + i\lambda^2\Big[a\ln\Lambda^2 - b(s_0) - b(t_0) - b(u_0)\Big] \tag{9}$$

Solving for the bare coupling:

$$\lambda = \lambda_R + \lambda_R^2\Big[a\ln\Lambda^2 - b(s_0) - b(t_0) - b(u_0)\Big] + \mathcal{O}(\lambda_R^3) \tag{10}$$

Now substitute this back into the amplitude at *arbitrary* kinematics:

The Renormalized Amplitude

$$i\mathcal{M} = -i\lambda_R - i\lambda_R^2\left[\,b(s) - b(s_0) + b(t) - b(t_0) + b(u) - b(u_0)\,\right] + \mathcal{O}(\lambda_R^3) \tag{11}$$

What Just Happened

The cutoff Λ has completely disappeared. The amplitude is expressed entirely in terms of:

- λ_R: the measured coupling at the reference point (an experimental input).
- $b(s) - b(s_0)$, etc.: *differences* of finite functions evaluated at different kinematics.

No infinities remain. The theory is predictive: given one measurement (λ_R), you can predict the scattering amplitude at any other energy.

4 Renormalization in QED: Vacuum Polarization

Now we apply the same logic to quantum electrodynamics, which is where Griffiths focuses in Section 7.9.

4.1 The tree-level amplitude

For electron-muon scattering at lowest order:

$$\mathcal{M}_0 = \frac{-ig_e^2}{q^2}\Big[\bar{u}(p_3)\gamma^\mu u(p_1)\Big]\Big[\bar{u}(p_4)\gamma_\mu u(p_2)\Big] \tag{12}$$

This is two fermion currents connected by a single photon propagator $-ig_{\mu\nu}/q^2$.

4.2 The vacuum polarization loop

The virtual photon can split into an e^+e^- pair, which then recombines. This gives a loop with two fermion propagators and an integral over the loop momentum k:

$$I_{\mu\nu} = -g_e^2 \int \frac{d^4k}{(2\pi)^4} \frac{\mathrm{Tr}\big[\gamma_\mu(\not{k}+mc)\gamma_\nu(\not{k}-\not{q}+mc)\big]}{(k^2-m^2c^2)\big[(k-q)^2-m^2c^2\big]} \tag{13}$$

The trace arises because the loop is a closed fermion line with no external spinors.

4.3 Tensor structure

After integrating over k, the result can only depend on q^μ and $g_{\mu\nu}$ (the only available tensorial objects). So:

$$I_{\mu\nu} = A(q^2)\, g_{\mu\nu} + B(q^2)\, q_\mu q_\nu \tag{14}$$

The $q_\mu q_\nu$ term vanishes when contracted with the external currents (because $q_\mu \bar{u}\gamma^\mu u = 0$ by the Dirac equation). So only the $g_{\mu\nu}$ piece matters physically. Griffiths writes it as:

$$I_{\mu\nu} = -ig_{\mu\nu}\, q^2\, I(q^2) + q_\mu q_\nu\, J(q^2) \tag{15}$$

4.4 The modified propagator

Including the loop amounts to replacing the photon propagator:

$$\frac{-ig_{\mu\nu}}{q^2} \;\longrightarrow\; \frac{-ig_{\mu\nu}}{q^2} - \frac{i}{q^4}\, I_{\mu\nu} \;=\; \frac{-ig_{\mu\nu}}{q^2}\Big[1 + I(q^2)\Big] \tag{16}$$

If you sum all orders of vacuum polarization insertions (one loop, two loops, ...), the geometric series gives:

$$\frac{-ig_{\mu\nu}}{q^2} \;\longrightarrow\; \frac{-ig_{\mu\nu}}{q^2\big[1 - I(q^2)\big]} \tag{17}$$

4.5 Evaluating $I(q^2)$: the divergence

Griffiths evaluates the integral and finds (Equation 7.186):

$$I(q^2) = \frac{1}{12\pi^2}\left[\ln\left(\frac{M^2}{m^2c^2}\right) - f\left(\frac{-q^2}{m^2c^2}\right)\right] \tag{18}$$

where M is a momentum cutoff, and $f(x)$ is a finite, calculable function with the limiting forms:

$$f(x) \approx \begin{cases} x/5 & (x \ll 1, \text{ nonrelativistic}) \\ \ln x & (x \gg 1, \text{ ultrarelativistic}) \end{cases} \tag{19}$$

The $\ln M$ diverges as $M \to \infty$. But it carries *no dependence on* q^2—it's a constant.

4.6 Defining the physical charge

Now apply the renormalization procedure. Define the physical (renormalized) coupling as:

$$g_R^2 \equiv g_e^2\left[1 + \frac{g_e^2}{12\pi^2}\ln\left(\frac{M^2}{m^2c^2}\right)\right] \tag{20}$$

This absorbs the divergent $\ln M$ into the definition of the measured charge. Rewriting the amplitude in terms of g_R:

Renormalized QED Amplitude (Griffiths Eq. 7.190)

$$\mathcal{M} = \frac{-g_R^2}{q^2}\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right]\left\{1 + \frac{g_R^2}{12\pi^2} f\left(\frac{-q^2}{m^2c^2}\right)\right\}\left[\bar{u}(p_4)\gamma_\mu u(p_2)\right] \tag{21}$$

The cutoff M is gone. What remains is a finite, q^2-dependent correction encoded in f.

5 The Running Coupling Constant

The Running Coupling

The finite leftover from renormalization tells us that the effective strength of the electromagnetic interaction depends on the momentum transfer q^2. The coupling "runs" with energy scale.

5.1 $\alpha(q^2)$ in QED

Absorbing the finite correction into the coupling itself:

$$\alpha(q^2) = \frac{\alpha(0)}{1 - \dfrac{\alpha(0)}{3\pi} f\left(\dfrac{-q^2}{m^2c^2}\right)} \tag{22}$$

where $\alpha(0) \approx 1/137$ is the value measured at $q^2 = 0$ (the classical, long-distance limit).

Since $f > 0$ for $q^2 < 0$ (spacelike momentum transfer, the physical scattering regime), the denominator is *less than 1*, so $\alpha(q^2) > \alpha(0)$. The coupling increases at shorter distances.

The Physical Picture: Vacuum Screening

The vacuum is filled with virtual e^+e^- pairs that act as tiny dipoles. They orient themselves to partially screen any electric charge—exactly like a dielectric medium. At large distances, you see the screened charge: $\alpha \approx 1/137$. As you probe at shorter distances (higher q^2), you penetrate the screening cloud and see more of the "bare" charge underneath. Hence α grows.

5.2 How big is the effect?

In nonrelativistic physics ($|q^2| \ll m^2c^2$), the function $f \approx -q^2/(5m^2c^2)$, which is tiny. Even in a head-on collision at the energy scale of the electron mass, the correction to α is only about 6×10^{-6}. That's why this effect was invisible to everyone from Coulomb to Millikan.

But at the scale of the Z boson ($q^2 \sim M_Z^2$):

$$\alpha(M_Z^2) \approx \frac{1}{128} \tag{23}$$

which is measurably different from 1/137. And the effect *has* been measured, confirming the prediction of vacuum polarization.

5.3 Contrast with QCD

In QCD, the story reverses. The gluon self-interaction (absent in QED since photons don't carry charge) contributes to the loop corrections with the *opposite sign*, overwhelming the quark-loop screening effect. The result:

$$\alpha_s(q^2) = \frac{\alpha_s(\mu^2)}{1 + \dfrac{\alpha_s(\mu^2)}{12\pi}(33 - 2n_f)\ln\left(\dfrac{q^2}{\mu^2}\right)} \tag{24}$$

where n_f is the number of active quark flavors. Since $33 - 2n_f > 0$ for $n_f \leq 16$ (and there are only 6 quarks), the coupling *decreases* at high q^2. This is **asymptotic freedom**: quarks interact weakly at short distances and strongly at large distances.

QED vs. QCD Running

	QED	**QCD**
Loop content	e^+e^- pairs	quark loops + **gluon loops**
Screening effect	Screening only	Screening + **anti-screening**
Net effect	α increases at high q^2	α_s **decreases** at high q^2
Physical picture	Vacuum dielectric	Asymptotic freedom
Perturbation theory	Works at all accessible q^2	Works only at high q^2

6 The Ward Identity: Why Charge Stays Universal

Vacuum polarization isn't the only fourth-order correction. There are also vertex corrections and self-energy diagrams that modify the electron's mass and magnetic moment.

The Worry

The vertex corrections involve the *mass* of the particle to which the virtual photon attaches. If these contributed to charge renormalization, the electron and muon would receive *different* charge corrections—they'd end up with different effective charges. That would be a disaster: it would violate the observed universality of electric charge.

The **Ward identity** guarantees this doesn't happen. It's a consequence of the local gauge invariance of QED ($U(1)$ symmetry), and it says: the vertex correction contributions to charge renormalization cancel each other exactly, to all orders in perturbation theory.

As a result, the *entire* charge renormalization comes from vacuum polarization (the photon propagator correction), which depends only on the photon—not on which particle it's coupling to. The renormalized charge is therefore the same for every charged particle, regardless of its mass.

This is not a coincidence. It's a *theorem* that follows from gauge invariance. And it illustrates a deep principle: **symmetries of the Lagrangian survive renormalization**.

7 What Makes a Theory Renormalizable?

Not every QFT can be renormalized. The key question is: as you include higher and higher-order diagrams, do you keep encountering *new types* of divergences, or can all divergences be absorbed into a *finite set* of redefined parameters?

Renormalizability

A theory is **renormalizable** if all divergences at every order can be absorbed by redefining a finite number of parameters (masses and coupling constants) already present in the Lagrangian. The number of counterterms you need is fixed and finite.

For QED, you need exactly **three** counterterms:

1. **Electron self-energy** Σ: renormalizes the electron mass.
2. **Photon self-energy** $\Pi^{\mu\nu}$ (vacuum polarization): renormalizes the charge.
3. **Vertex correction** Γ^{μ}: renormalizes the coupling (but its charge-renormalization piece cancels by the Ward identity, leaving only the anomalous magnetic moment).

These three corrections, plus their counterterms, are sufficient to render *all* QED predictions finite, to *all* orders. This was proven by Dyson in 1949 and is the reason QED is the most precisely tested theory in physics.

7.1 Power counting: a quick diagnostic

There's a simple rule of thumb. In 4 spacetime dimensions, a coupling constant with mass dimension $[\lambda] = M^d$ contributes to divergences as follows:

- $d > 0$: **super-renormalizable**. Only a finite number of diagrams diverge.
- $d = 0$: **renormalizable**. Divergences occur at every order, but can all be absorbed into a finite set of counterterms. (QED, QCD, ϕ^4 theory, the Standard Model.)

- $d < 0$: **non-renormalizable**. Each new order introduces new types of divergences requiring new counterterms. The theory loses predictive power at high energies. (Fermi's theory of weak interactions, general relativity.)

Non-Renormalizable $\neq$ Useless

A non-renormalizable theory can still be an excellent *effective field theory* at low energies. Fermi's theory of weak interactions (G_F) works beautifully for low-energy beta decay. It only fails at energies around $M_W \sim 80$ GeV, where it must be replaced by the full electroweak theory (which *is* renormalizable). Similarly, general relativity works as a quantum effective field theory at energies far below the Planck scale.

8 The Renormalization Group: The Bigger Picture

The Renormalization Group

The renormalization group (RG) is the framework that formalizes how the parameters of a theory change with the energy scale at which you observe the physics. It turns the "running coupling" from a one-loop curiosity into a powerful organizing principle for all of physics.

8.1 The β-function

The rate at which a coupling constant changes with energy scale μ is governed by the β**-function**:

$$\mu \frac{d\lambda}{d\mu} = \beta(\lambda) \tag{25}$$

This is a differential equation: given λ at one scale, integrate to find λ at any other scale.

For QED at one-loop order:

$$\beta_{\text{QED}}(e) = \frac{e^3}{12\pi^2} > 0 \tag{26}$$

Positive β means the coupling *grows* in the UV (high energy). This is the QED screening we've been discussing.

For QCD at one-loop order:

$$\beta_{\text{QCD}}(g_s) = -\frac{g_s^3}{16\pi^2}(11 - \tfrac{2}{3}n_f) \tag{27}$$

Negative β (for $n_f < 17$) means the coupling *shrinks* in the UV. This is asymptotic freedom.

8.2 Fixed points

If $\beta(\lambda^*) = 0$ for some value λ^*, the coupling doesn't run at that point—it's a **fixed point**. The theory becomes scale-invariant there. Fixed points are central to the theory of phase transitions and critical phenomena, where the renormalization group finds its most powerful applications.

9 Summary: The Logic of Renormalization

The Logical Chain

1. Loop diagrams produce divergent integrals.

2. The divergences come from the large-k (short-distance) region and are **independent of external kinematics**.

3. Because the divergences are kinematic-independent, they can be absorbed into redefinitions of the Lagrangian parameters (mass, coupling constant).

4. The parameters we started with ("bare" parameters) were never observable. What we measure in the lab are the "dressed" (renormalized) parameters, which already include all quantum corrections.

5. After expressing everything in terms of renormalized parameters, all predictions are finite. The leftover finite corrections depend on q^2, giving the running coupling constant.

6. The Ward identity (gauge invariance) guarantees that charge renormalization is universal—the same for all charged particles.

7. A theory is renormalizable if this procedure works to all orders with only finitely many counterterms. QED, QCD, and the full Standard Model are all renormalizable.

The Deepest Lesson

Renormalization isn't a band-aid for a broken theory. It's the recognition that quantum field theories describe physics at *different scales differently*, and the parameters of the theory must be understood as scale-dependent quantities. The "constants" of nature aren't constant—they run. The renormalization group is the language in which nature tells us how physics at one scale is connected to physics at another.